

5.2 Energy: GPE & KE

Question Paper

Course	CIE A Level Physics
Section	5. Work, Energy & Power
Topic	5.2 Energy: GPE & KE
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A loaded aeroplane has a total mass of $1.2 \times 10^5 \text{ kg}$ while climbing after take-off. It climbs at an angle of 23° to the horizontal with a speed of 50 m s^{-1} .

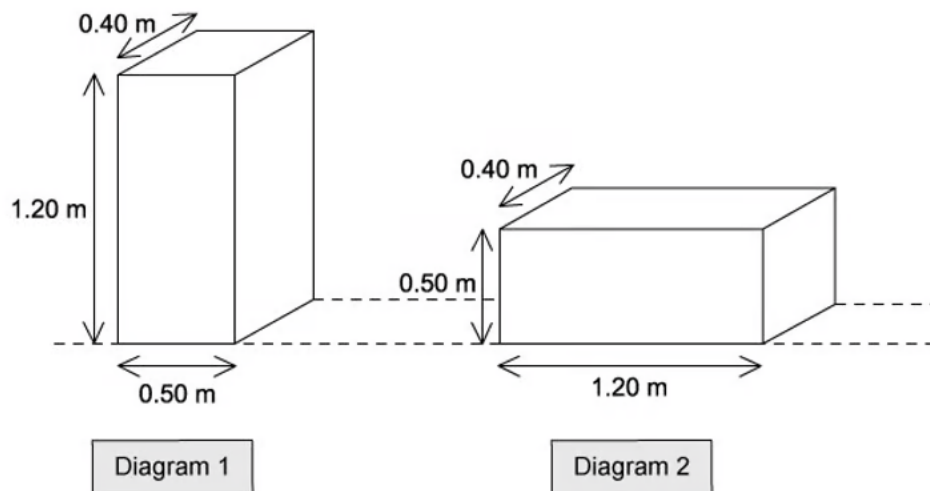
What is the rate at which it is gaining potential energy at this time?

- A. $2.3 \times 10^6 \text{ J s}^{-1}$
- B. $2.5 \times 10^6 \text{ J s}^{-1}$
- C. $2.3 \times 10^7 \text{ J s}^{-1}$
- D. $2.5 \times 10^7 \text{ J s}^{-1}$

[1 mark]

Question 2

A uniform solid cuboid of concrete of dimensions $0.50 \text{ m} \times 1.20 \text{ m} \times 0.40 \text{ m}$ and weight 4000 N rests on a flat surface with the 1.20 m edge vertical as shown in diagram 1.



What is the minimum energy required to roll the cuboid through 90° to the position shown in diagram 2 with the 0.50 m edge vertical?

- A. 200 J
- B. 400 J
- C. 1400 J
- D. 2600 J

[1 mark]

Question 3

A steel sphere is dropped vertically onto a horizontal metal plate. The sphere hits the plate with a speed u , leaves it at a speed v , and rebounds vertically to half of its original height.

Which expression gives the value of $\frac{v}{u}$?

A. $\frac{1}{2^2}$

B. $\frac{1}{2}$

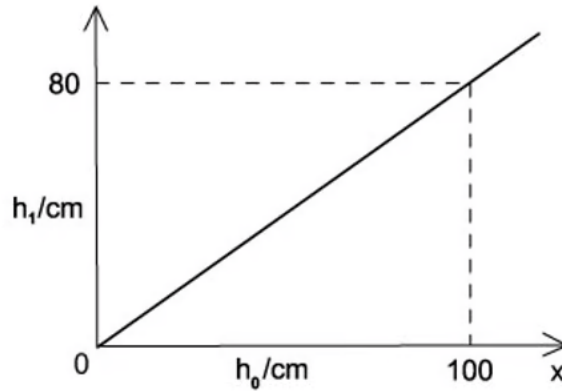
C. $\frac{1}{\sqrt{2}}$

D. $1 - \frac{1}{\sqrt{2}}$

[1 mark]

Question 4

A ball released from a height h_0 above a horizontal surface rebounds to a height h_1 after one bounce. The graph that relates h_0 to h_1 is shown below.



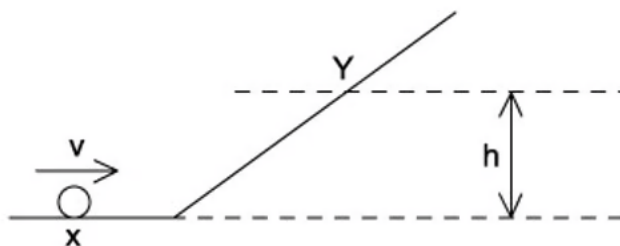
If the ball (of mass m) was dropped from an initial height h and made three bounces, the kinetic energy of the ball immediately after the third impact with the surface was

- A. $(0.8)^3 mgh$
- B. $(0.8)^2 mgh$
- C. $[1 - (3 \times 0.2)] mgh$
- D. $[1 - (0.8)^3] mgh$

[1 mark]

Question 5

An object of mass m passes a point X with a velocity v and slides up a frictionless incline to stop at point Y which is at a height h above X



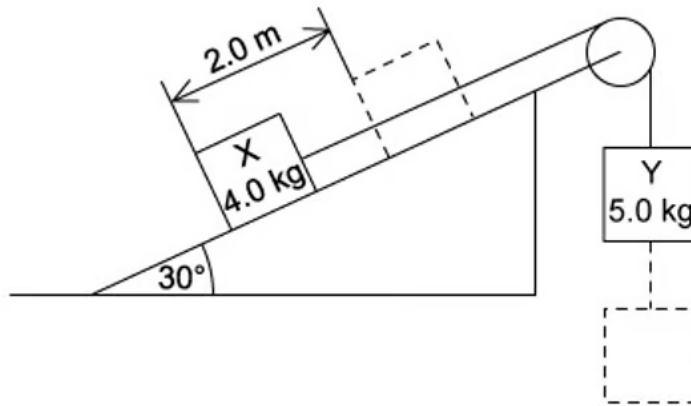
A second object of mass $\frac{1}{2}m$ passes X with a velocity of $\frac{1}{2}v$. To what height will it rise?

- A. $\frac{h}{4}$
- B. $\frac{h}{2}$
- C. $\frac{h}{\sqrt{2}}$
- D. $h\sqrt{2}$

[1 mark]

Question 6

The diagram shows two bodies X and Y connected by a light cord passing over a light, free-running pulley. X starts from rest and moves on a smooth plane inclined at 30° to the horizontal.



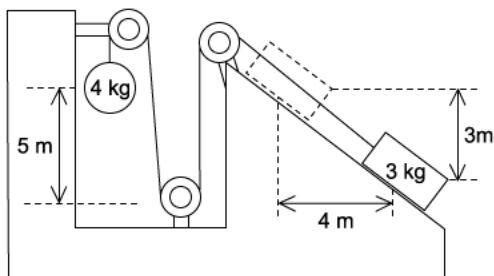
What will be the total kinetic energy of the system when X has travelled 2.0 m along the plane?

- A. 20 J
- B. 59 J
- C. 64 J
- D. 132 J

[1 mark]

Question 7

A 4 kg ball and a 3 kg block of wood were placed in an arrangement as shown below. When the ball was released, it fell through a distance of 5 m, pulling the block of wood up the ramp.



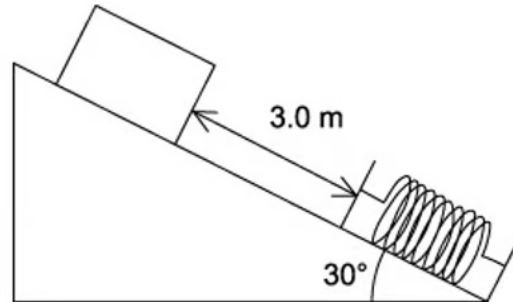
If the pulleys were well oiled and the surface of the ramp exerts a frictional force of 2 N, what is the final speed of the sphere after falling through the 5 m?

- A. 5.0 m s^{-1}
- B. 5.3 m s^{-1}
- C. 6.9 m s^{-1}
- D. 10.0 m s^{-1}

[1 mark]

Question 8

A block of mass 12 kg is released from rest and it slides down a 30° frictionless incline plane. It is stopped by a spring with a spring constant k as shown below.



The initial distance between the point of release and the uncompressed spring is 3.0 m . After the block comes to rest momentarily, the spring is compressed by 1.3 m .

What is the value of k ?

- A. 200 N m^{-1}
- B. 250 N m^{-1}
- C. 300 N m^{-1}
- D. 350 N m^{-1}

[1 mark]

Question 9

A falling mass pulls a trolley from rest on a horizontal table. Given that:

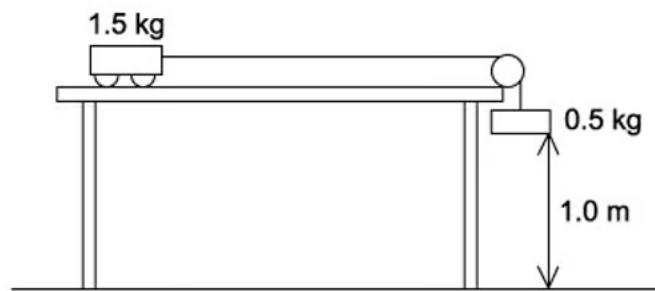
The mass of the trolley = 1.5 kg

The mass of the falling mass = 0.5 kg

The distance the mass falls through = 1.0 m

Frictional force from the table = 0.8 N

The distance the trolley moves = 1.2 m



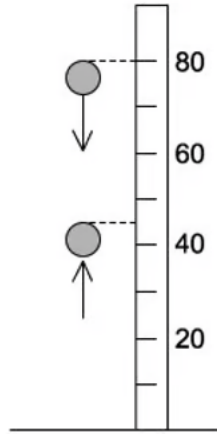
What is the maximum kinetic energy of the trolley?

- A. 3.0 J
- B. 4.9 J
- C. 15 J
- D. 20 J

[1 mark]

Question 10

A solid rubber ball has a diameter of 8.0 cm. It is released from rest with the top of the ball 80 cm above a horizontal surface. It falls vertically and then bounces back up so that the maximum height reached by the top of the ball is 45 cm, as shown.



If the kinetic energy of the ball is 0.75 J just before it strikes the surface, what is its kinetic energy just after it leaves the surface?

- A. 0.36 J
- B. 0.39 J
- C. 0.40 J
- D. 0.42 J

[1 mark]

Model Answers: Hard

1

The correct answer is **C** because:

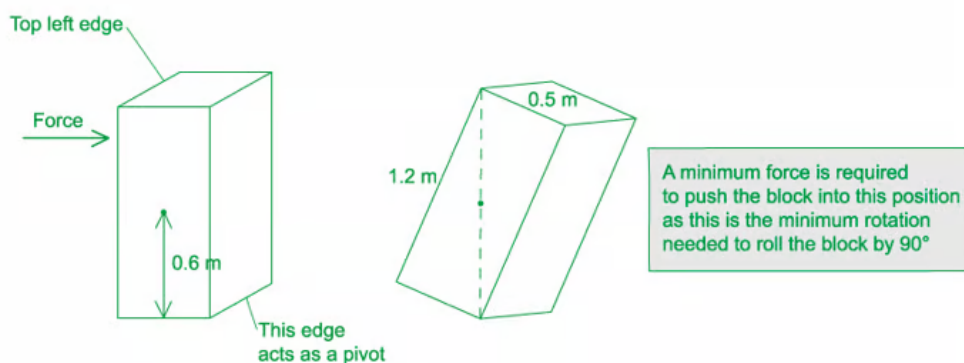
- Rate of gaining potential energy = $\frac{\text{gain in GPE}}{\text{time}}$
- $\text{GPE} = mgh$
 - Where h is the vertical height
- So, to consider the gain in GPE, we need to consider the vertical component of speed
- The aeroplane climbs at a speed of 50 m at an angle of 23° to the horizontal
 - Vertical component of speed = $50 \sin 23^\circ$
- Rate of gaining potential energy = $\frac{\text{gain in GPE}}{\text{time}}$
- Rate of gaining potential energy = $\frac{mgh}{t} = mg\left(\frac{h}{t}\right)$
 - Where $\frac{h}{t}$ is the vertical speed at which the aeroplane climbs
- Rate of gaining potential energy = $(1.2 \times 10^5) \times 9.81 \times 50 \sin 23^\circ$
- Rate of gaining potential energy = $2.3 \times 10^7 \text{ J s}^{-1}$

2

The correct answer is **A** because:

- Since the solid cuboid is uniform, its centre of mass may be considered to be at the centre of the cuboid, that is, initially at a height of 0.6 m
 - Weight = $mg = 4000 \text{ N}$
- Therefore, before the force is applied, the GPE of the cuboid is equal to
 - $mgh = 4000 \times 0.6$
- As illustrated by the diagram, as the cuboid rolls about the bottom right edge, the centre of mass of the cuboid rises to a greater height, so, its potential energy increases and reaches a maximum and afterwards, the cuboid will fall under gravity

- So, the minimum energy is equal to the change in energy from these two positions
- Consider the position for maximum potential energy:
 - The centre of mass will be at the middle of the dotted line and this forms a right-angle triangle as shown in the diagram
- Using Pythagoras' theorem:
 - Vertical dotted line = $\sqrt{0.5^2 + 1.2^2} = 1.3 \text{ m}$
- Height of centre of mass in new position = $\frac{1.3}{2} = 0.65 \text{ m}$
- GPE of cuboid in new position :
 - $mgh = 4000 \times 0.65$
- Minimum energy required to roll cuboid = Rise in potential energy
- Minimum energy = $(4000 \times 0.65) - (4000 \times 0.6) = 200 \text{ J}$



3

The correct answer is **C** because:

- Let the height from which the sphere is dropped = h
- As the sphere falls, its gravitational potential energy gets converted into kinetic energy.
- Its speed when it hits the plate is u , so:
 - $mgh = \frac{1}{2} mu^2$
 - Speed $u = \sqrt{2gh}$
- As the sphere rebounds and move vertically up, its kinetic energy gets converted into gravitational potential energy

- The height is now half the height where it has dropped initially, so:

$$\circ \frac{1}{2}mv^2 = mg\left(\frac{h}{2}\right)$$

$$\circ \text{Speed } v = \sqrt{gh}$$

- So, the ratio of $\frac{v}{u}$ is:

$$\circ \text{Ratio} = \frac{v}{u} = \frac{\sqrt{gh}}{\sqrt{2gh}}$$

$$\circ \text{Ratio} = \frac{v}{u} = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{gh}}{\sqrt{gh}}$$

$$\circ \text{Ratio} = \frac{v}{u} = \frac{1}{\sqrt{2}}$$

4

The correct answer is **A** because:

- Initially, the potential energy of ball = mgh
 - h_0 is the height before release and h_1 is the height after one bounce
- Since the graph is a straight line, the ratio of h_1 / h_0 (gradient) is constant
 - Gradient = $\frac{\Delta h_1}{\Delta h_0} = \left(\frac{80 - 0}{100 - 0} \right) = 0.8$
- So, after one bounce the potential energy becomes 0.8 of the initial potential energy
- The kinetic energy of the ball immediately after an impact with the surface is the same as the potential energy at maximum height as all kinetic energy is converted into potential energy
 - After 1st bounce, kinetic energy = $0.8 mgh$
 - After 2nd bounce, kinetic energy = $0.8 \times (0.8mgh) = (0.8)^2 mgh$
 - After 3rd bounce, kinetic energy = $(0.8)^3 mgh$

5

The correct answer is **A** because:

- For the object of mass m and velocity v :

- At point X, Kinetic energy $= \frac{1}{2}mv^2$
- From conservation of energy :
- Potential energy at point Y = Kinetic energy at point X
 - $mgh = \frac{1}{2}mv^2$
 - Height, $h = \frac{v^2}{2g}$
- For the object of mass $\frac{m}{2}$ and velocity $\frac{v}{2}$:
 - At point X, Kinetic energy $= \frac{1}{2}\left(\frac{m}{2}\right)\left(\frac{v}{2}\right)^2 = \frac{mv^2}{16}$
- Potential energy at point Y = Kinetic energy at point X
- Let the height at point Y = x
 - $\left(\frac{m}{2}\right)gx = \frac{mv^2}{16}$
 - Height, $x = \frac{v^2}{8g} = \frac{1}{4}\left(\frac{v^2}{2g}\right) = \frac{h}{4}$

6

The correct answer is **B** because:

- The acceleration due to gravity of 9.8 m s^{-2} acts downwards on Y since it is hanging freely
 - Weight of Y $= mg = 5.0 \times 9.8 = 49 \text{ N}$
- As X is attached to Y, it will move up the slope as Y falls
- So, the acceleration in the whole system is therefore **not** 9.8 m s^{-2}
- We need to find the resultant acceleration which acts on the total mass in the system. The component of the weight of X will act on the system:
 - $mg \sin 30 = 4.0 \times 9.8 \times \sin 30 = 19.6 \text{ N}$
- The weight of Y contributes a force up the slope while the component of weight of X contributes a force down the slope
- Resultant force in system $= ma = 49 - 19.6 = 29.4 \text{ N}$
 - $(4.0 + 5.0)a = 29.4 \text{ N}$
 - Acceleration, $a = \frac{29.4}{9} = 3.27 \text{ m s}^{-2}$
- This is the resultant acceleration on each X and Y.

- After 3rd bounce, kinetic energy = $(0.8)^3 mgh$

5

The correct answer is **A** because:

- For the object of mass m and velocity v :
 - At point X, Kinetic energy = $\frac{1}{2}mv^2$
- From conservation of energy :
- Potential energy at point Y = Kinetic energy at point X
 - $mgh = \frac{1}{2}mv^2$
 - Height, $h = \frac{v^2}{2g}$
- For the object of mass $\frac{m}{2}$ and velocity $\frac{v}{2}$:
 - At point X, Kinetic energy = $\frac{1}{2}\left(\frac{m}{2}\right)\left(\frac{v}{2}\right)^2 = \frac{mv^2}{16}$
- Potential energy at point Y = Kinetic energy at point X
- Let the height at point Y = x
 - $\left(\frac{m}{2}\right)gx = \frac{mv^2}{16}$
 - Height, $x = \frac{v^2}{8g} = \frac{1}{4}\left(\frac{v^2}{2g}\right) = \frac{h}{4}$

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- The acceleration due to gravity of 9.8 m s^{-2} acts downwards on Y since it is hanging freely
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- The weight of Y contributes a force up the slope while the component of weight

of X contributes a force down the slope

- Resultant force in system = $ma = 49 - 19.6 = 29.4 \text{ N}$
 - $(4.0 + 5.0)a = 29.4 \text{ N}$
 - Acceleration, $a = \frac{29.4}{9} = 3.27 \text{ m s}^{-2}$
- This is the resultant acceleration on each X and Y.
- Initially both X and Y are at rest. Consider the equation for uniformly accelerated motion:
 - $v^2 = u^2 + 2as$ where $u = 0$, $v = ?$, $s = 2.0 \text{ m}$ and $a = 3.27 \text{ m s}^{-2}$
 - $v^2 = 0 + 2(3.27)(2)$
 - $v = \sqrt{2 \times 3.27 \times 2} = 3.62 \text{ m s}^{-1}$
- Kinetic energy of X = $\frac{1}{2}(4.0)(3.62)^2$
- Kinetic energy of Y = $\frac{1}{2}(5.0)(3.62)^2$
- Total kinetic energy of system = $\frac{1}{2}(4.0)(3.62)^2 + \frac{1}{2}(5.0)(3.62)^2 = 58.96 \text{ J}$
 - Therefore, the total kinetic energy is 59 J

7

The correct answer is **B** because:

- From the conservation of energy:
 - Change in GPE = Change in KE + Work done against friction
- In equation form we can write this such that:
 - $\Delta GPE = \Delta KE + Fd$
- The change in GPE comes from the loss in GPE from the 4 kg mass and the gain in GPE from the 3 kg mass:
 - $\Delta GPE = mgh = (4 \times 9.8 \times 5) - (3 \times 9.8 \times 3) = 107.8 \text{ J}$
- The change in KE comes from the gain in GPE from the 4 kg mass and the 3 kg mass, the masses are connected so both travel with speed v :
 - $\Delta KE = \frac{1}{2}mv^2 = \frac{1}{2}(3)v^2 + \frac{1}{2}(4)v^2 = \frac{7v^2}{2}$
- The work done against friction is given by $Fd = 2 \times 5 = 10 \text{ J}$
 - $107.8 = \frac{7v^2}{2} + 10$

- $v^2 = \frac{2}{7}(107.8 - 10) = 27.9$
- $v = \sqrt{27.9} = 5.29 \text{ m s}^{-1}$
- Therefore, the final speed of the mass is 5.3 m s^{-1}

8

The correct answer is **C** because:

- From the conservation of energy:
 - Change in GPE = Change in KE + Work done against friction
- We are told the slope is frictionless, so we can disregard the work done against friction as it is negligible
- Change in GPE as the block slides down the slope and compresses the spring is
 - $\Delta GPE = mgh = (12)(9.8)(3 + 1.3)\sin 30 = 252.84 \text{ J}$
- The change in KE is equal to the elastic energy stored by the spring when it is compressed, so:
 - $\Delta KE = \frac{1}{2}kx^2 = \frac{1}{2}k(1.3)^2$
- Where k is the spring constant and x is the compression of the spring
 - $\Delta GPE = \Delta KE$
 - $252.84 = \frac{1}{2}k(1.3)^2$
 - $k = \frac{2 \times 252.84}{(1.3)^2} = 299.2 \text{ N m}^{-1}$
- Therefore, the spring constant is 300 N m^{-1}

9

The correct answer is **A** because:

- From the conservation of energy:
 - Loss in PE of the falling mass = Gain in KE of the whole system + Work done against friction
- Loss in PE is equal to $= m_m gh = 0.5 \times 9.8 \times 1 = 4.9 \text{ J}$
- Total gain in KE is equal to $= \frac{1}{2}(m_t + m_m)v^2 = \frac{1}{2}(1.5 + 0.5)v^2 = v^2$
 - Where m_t and m_m are the masses of the trolley and falling mass respectively

- Work done against friction = $Fd = 0.8 \times 1.2 = 0.96J$
- The final speed of the trolley and mass is:
 - $4.9 = v^2 + 0.96$
 - $v = \sqrt{4.9 - 0.96} = 1.98 \text{ m s}^{-1}$
- So, the maximum kinetic energy of the trolley is equal to
 - $\frac{1}{2} m_t v^2 = \frac{1}{2} (1.5)(1.98)^2 = 2.955J$
- Therefore, the maximum kinetic energy of the trolley is 3.0 J

10

The correct answer is **B** because:

- Let's first consider the bottom of the ball at 72 cm at the start and 37 cm at the end
- From the conservation of energy, as the ball falls, all the potential energy (PE) is being converted to kinetic energy
- Just before it strikes the surface, the kinetic energy is 0.75 J as all PE has been converted to KE
- At this point, the bottom of the ball has zero PE and a KE of 0.75 J
 - This value of KE is the value at the bottom of the ball, so, throughout our calculation we need to consider the bottom of the ball instead of the top
- PE of bottom of the ball at the point of release = KE of bottom of ball just before striking the surface
 - $\Delta GPE = mgh = \Delta KE$
 - $m \times 9.81 \times 0.72 = 0.75$
 - Mass of the ball, $m = \frac{0.75}{9.81 \times 0.72} = 0.1062 \text{ Kg}$
- Now, for the rebound, all of the kinetic energy just after the ball leaves the surface is converted to potential energy at the maximum height reached
- The maximum of height of the bottom of the ball after rebound is $45 - 8 = 37 \text{ cm}$
- KE just after leaving the surface = PE of bottom of ball at a height of 37 cm
 - $mgh = \Delta KE$
 - $KE = 0.1062 \times 9.81 \times 0.37 = 0.3855J$
- Therefore, the kinetic energy just after the ball leaves the surface is 0.39 J
 - Note that 0.75 J is the KE of the ball just before hitting the surface while 0.39 J is the KE of the ball just after leaving the surface. This indicates that some

energy $(0.75 - 0.39)$ J has been lost

